

Module 1 Annotation table

Source & page	Claim made by the authors	Evidence or support cited	Questions it raises for my work
Bommasani et al, p. 3	“A foundation model is any model that is trained on broad data (generally using self-supervision at scale) that can be adapted (e.g., fine-tuned) to a wide range of downstream tasks..”	BERT [Devlin et al. 2019], GPT-3 [Brown et al. 2020], and CLIP [Radford et al. 2021]. Bender et al. 2021	What is BERT and CLIP. BERT [Bidirectional Encoder Representations from Transformers] = “machine learning (ML) framework for natural language processing. In 2018, Google developed this algorithm to improve contextual understanding of unlabeled text across a broad range of tasks by learning to predict text that might come before and after (bi-directional) other text.” ~ BERT is used for a wide variety of language tasks. Below are examples of what the framework can help you do: • Determine if a movie’s reviews are positive or negative • Help chatbots answer questions • Help predicts text when writing an email • Can quickly summarize long legal contracts • Differentiate words that have multiple meanings based on the surrounding text Source: https://h2o.ai/wiki/bert/#:~:text=Yes.%20BERT%20is%20a%20neural%2Dnetwork%2Dbased,of%20words%20in%20search%20queries. CLIP (<i>Contrastive Language–Image Pre-training</i>) builds on a large body of work on zero-shot transfer, natural language supervision, and multimodal learning. The idea of zero-data learning dates back over a decade⁸ but until recently was mostly studied in computer vision as a way of generalizing to unseen object categories.

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			Source: https://openai.com/index/clip/
Bommasani et al, p. 3	Foundation models emphasize emergence and homogenization. Emergence = the behavior of a system is implicitly induced rather than explicitly constructed and can have unanticipated consequences. Homogenization = consolidation of methods for building ML systems across a wide range of applications Opportunities: Data sources from multiple stakeholders and using a variety of data source modalities (p. 54).	Not stated early on in first paragraphs 3.1 Healthcare and biomedicine Authors: Michihiro Yasunaga, Jing Huang, Camilo Ruiz, Yuhui Zhang, Giray Ogut, Saahil Jain, William Wang, Yusuf Roohani, Hongyu Ren, Antoine Bosselut, Ehsan Adeli, Jure Leskovec, Russ Altman	Emergence is how the AI system behaves and responds is variable versus being intentionally constructed to respond in a well-defined, prescriptive manner. Because of this, how tasks are ‘learned’ and performed can produce unintended consequences can occur and cannot be anticipated. This is a huge caution for healthcare. How are clinical AI platforms constructed and how are they emerging/homogenizing?

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	Risk: can create points of failure across models		
Bomma sani et al, p. 4	Prediction capabilities emerge – example in-context learning		
Bomma sani et al, p. 5-6	In-context learning allows for adapting a LM for a specific task without being specifically trained. This adaptation allows for scale without having to start a LM from scratch	Brown et al. 2020 Collobert and Weston [2008] Example also on p. 54	What are healthcare in-context learning examples?
Bomma sani et al, p. 5-7	Training approaches: ▪ Transfer learning ▪ Self-supervised learning ▪ Machine learning ▪ Deep learning ▪ In-context learning	See image p. 6 for data sources	

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	People as ‘ultimate source’ of data into training a foundational model. Include: Data, images, speech, structured data and 3D signals		
Bommasani et al, p. 7	“foundation” to connote the significance of architectural stability, safety, and security:	Definition by authors	Who / what checks the architecture in a model etc.?
Bommasani et al, p. 9	“A foundation model (e.g., Codex) can also be trained with another model (e.g., GPT-3) as a starting point.” Adaptation – Creating new model based on an original foundation model to perform a specific task		What are healthcare examples?
Bommasani et al, p. 12, 14	Report has four parts: 1. Capabilities 2. Applications 3. Technology	Percy Liang initiated and conceptualized the framing	

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	4. Society	and structure of the overall report. Liang and Rishi Bommasani - writing efforts etc. p. 14 (author contributions)	
Bommasani et al, p. 14	5 capabilities that can power applications 1. Language 2. Vision 3. Robotics 4. Reasoning and search 5. Interaction with users	Vision - Russakovsky et al. 2015, Radford et al. 2021 Reasoning/search - [Lakoff and Núñez 2000	
Bommasani et al, p. 15-16	Many applications, paper focuses on 3: 1. Healthcare Biomedicine 2. Law		

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	3. Education Healthcare – use data across multiple specialties and modalities Opportunities Value – efficiency and expertise Interaction between provider and AI and patient and AI Drug discovery Open ended research problems Risk: Bias in datasets and trials, privacy, regulations		
Bommasani et al, p. 17	Training – how models learn and obtain capabilities from training data “principled selection derived from systematic evidence	Devlin et al. 2019 Chen et al. 2020c	

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	and evaluation (§4.4: evaluation), and domain-generalty to provide rich, scalable, and unified training signal across data sources and modalities” Adaptation: Foundation models are ‘unfinished’ ! Must be adapted to complete tasks to reduce deficiencies of model		
Bommasani et al, p. 19	Foundation models must be reliable and safe so they are not used for negative goals		
Stanford HAI, 2026, p. 9	US vs China AI gap closed US has most data centers! 5,427	Not cited	DeepSeek is open-source AI reasoning model developed by the Chinese artificial intelligence company DeepSeek Anthropic - American AI safety and research company headquartered in San Francisco. Founded in 2021 by former OpenAI researchers developing t Claude

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	Responsible AI is an issue – safety issues are rising		
Stanford HAI, 2026, p. 9	Consumers using free AI – fast AI adoption U.S. AI adoption ranks 24th at 28.3%		
Stanford HAI, 2026, p. 10	Increased training emissions and water use	Grok 4's	
Stanford HAI, 2026, p. 10	Evidence limited to how AI is transforming clinical care Adoptions for clinical notes 83% less writing and reduced burnout Clinical AI studies relies on exam questions vs actual clinical data!	Not cited	

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	5% using real clinical data !		
Stanford HAI, 2026, p. 11	Formal education is lagging behind AI!	Not cited	Importance to teach
Stanford HAI, 2026, p. 173	Corporate investment in AI more than doubled in 2025 (globally) Generative AI used in at least one business function		
Stanford HAI, 2026, p. 186-187	AZ – 504 M private investment in AI Most investment in infrastructure/models/research/governance	Quid 2025 Chart 2026 Ai Index report	
Stanford HAI, 2026, p. 373-374	Higher workplace adoption than public adoption Predict 8% of US work hours in 2027	Forecasting Research Institute 2026	

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	18% in 2030		
Stanford HAI, 2026, p. 292	AI-relevant majors into two categories: AI software and AI hardware U of A not in the top list with AI degrees!	White House AI Talent Report 2025	
Stanford HAI, 2026, p. 303-304	University students' Gen AI uses by University students 2025 Understanding concept or subject (56%) Researching assignments and projects (52%) Generating initial ideas or first drafts (46%) Writing/editing (41%) -----	Chegg Global Student Survey, 2025 Chart: 2026 AI Index report	

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	Students who do not use AI tools are concerned about academic integrity issues, content accuracy and school policy restricting its use		